

Integration of Random Forest and Shadow Post-Processing for Optimized Land Cover Classification in GEE

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1. INTRODUCTION

Accurate land cover classification is essential for urban hydrological modeling and flood risk prediction, as impervious surfaces heavily influence runoff and peak flows [1]. To optimize the mapping of complex urban environments, this study implements Random Forest (RF) within Google Earth Engine (GEE), a machine learning approach that surpasses traditional parametric methods [2]. Additionally, spatial filters like focal mode are integrated to effectively reduce noise in densely built-up areas [3].

Objective: To present a geospatial and machine learning approach in GEE utilizing a high-resolution image in the Mburicao Stream Lower Basin.

2. METHODOLOGY

Study area: The basin of the Mburicao stream, in Asunción, Paraguay, is a dense urban area vulnerable to flash floods due to the increase in impervious surfaces.

The process for accurate land cover classification was carried out following the scheme presented in Figure 1.

Data: high-resolution image (1 m) y OSM masking delimited to the study area. A training dataset was constructed by collecting representative samples of five cover classes: Vegetation, Urban, Water, Roads, and Bare soil.

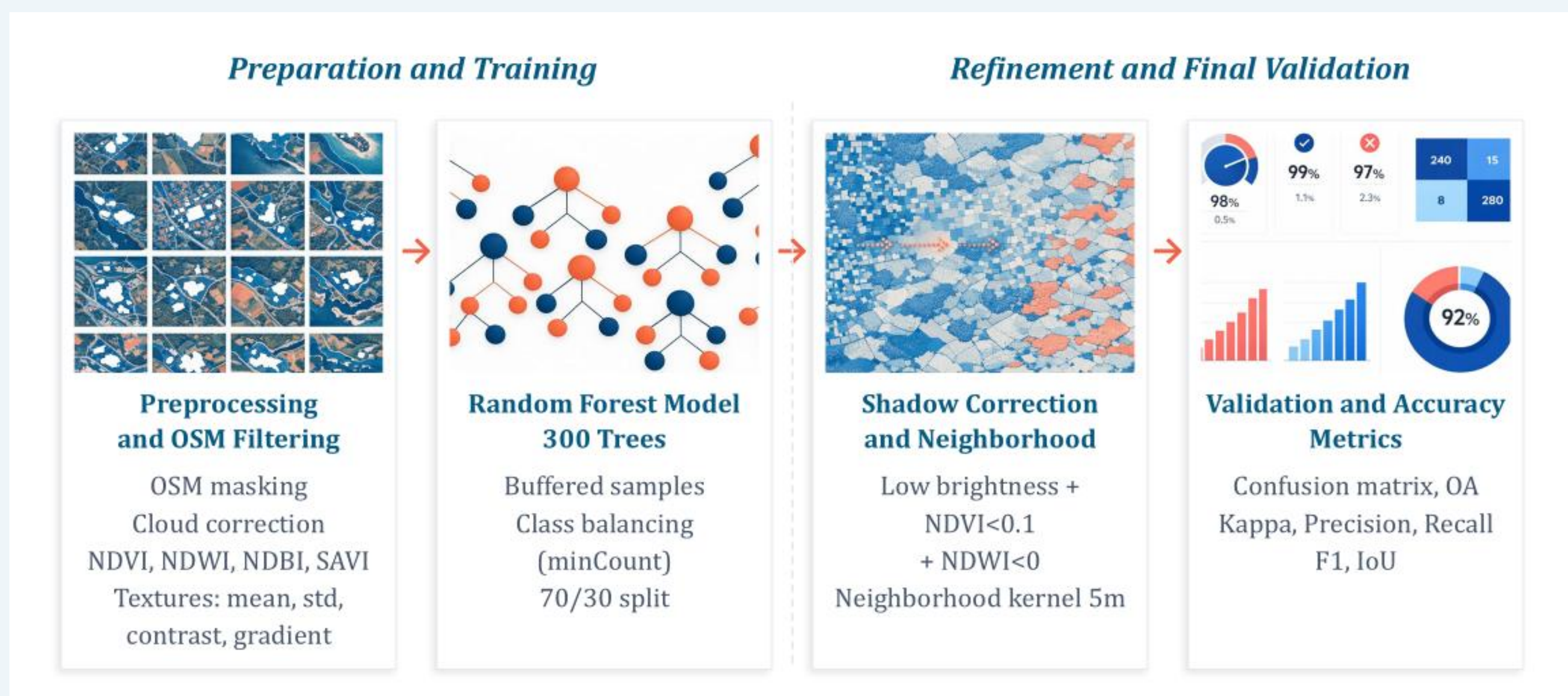


Figure 1: Workflow Satellite Image Classification and Refinement

3. RESULTS

The high overall accuracy (97.17%) and Kappa value (0.7390) confirm the robustness of the Random Forest classifier.

Table 1: Class Metrics with Performance Evaluation

Class	Precision (%)	Recall (%)	F1-Score	IoU (%)
Vegetation	99.08	99.48	99.28	98.57
Urban	92.45	67.67	78.14	64.12
Bare soil	78.70	58.40	67.04	50.43
Roads	27.81	71.65	40.07	25.05
Water	6.61	4.13	5.08	2.61

Class-wise performance is excellent for Vegetation (F1: 99.28), good for Urban (78.14), and acceptable for Bare soil (67.04). Roads (27.81% precision, overestimation) and Water (F1: 5.08, shadow-water confusion) are the problematic classes (Table 1).

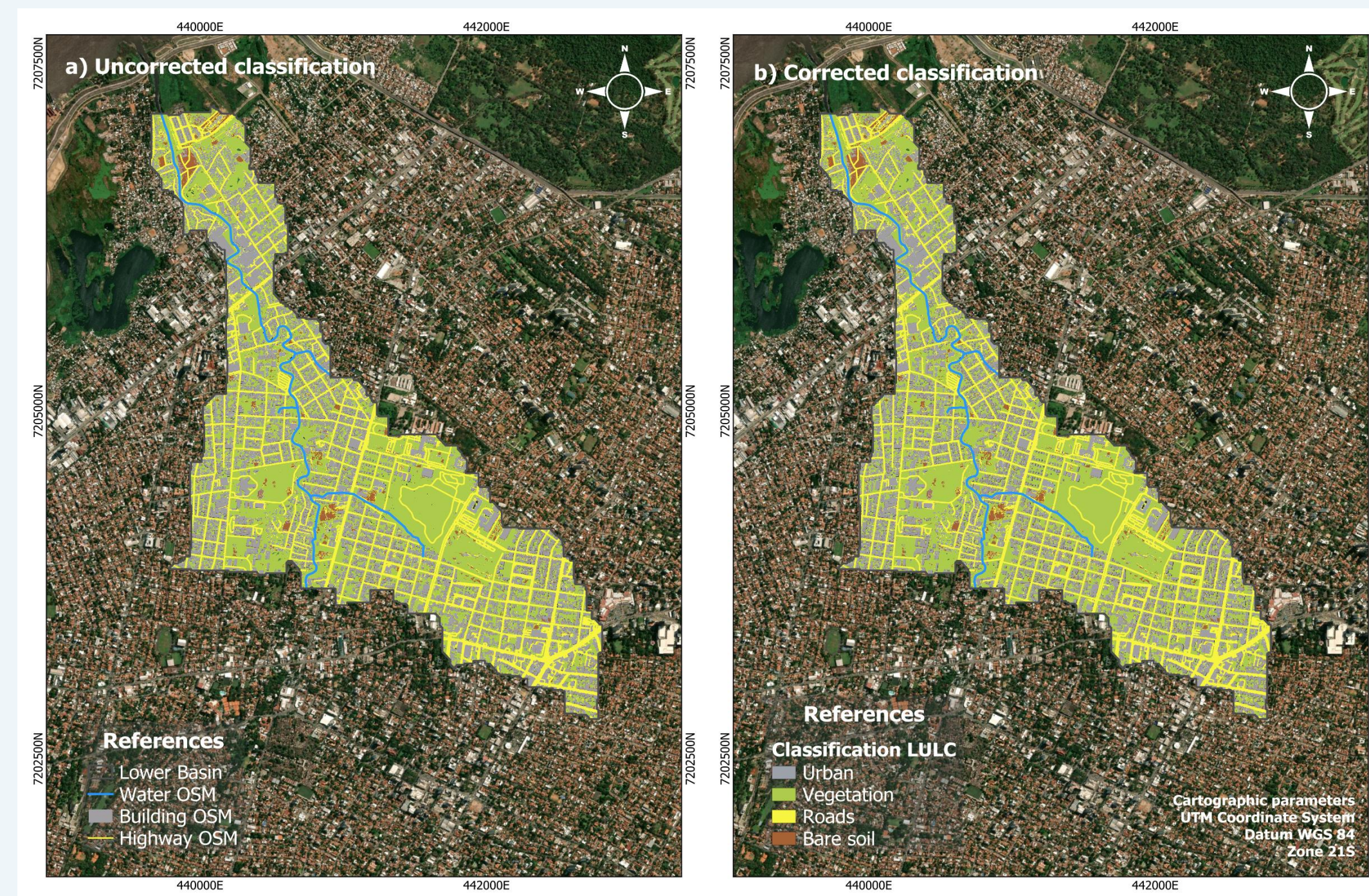


Figure 2: Map comparison before and after shadow correction:

Figure 2 demonstrates that the shadow correction did indeed modify the map spatially.

Table 2: Comparison of Areas by Class (ha)

Class	Uncorrected (ha)	Corrected (ha)	Variation (ha)	Variation (%)
Water	1.47	0.26	-1.21	-82.3%
Urban	71.96	72.57	+0.61	+0.8%
Vegetation	218.49	219.10	+0.61	+0.3%
Roads	53.65	54.26	+0.61	+1.1%
Bare soil	18.26	17.64	-0.62	-3.4%
TOTAL	363.83	363.83	0.00	0.0%

Shadow post-processing reduced water from 1.47 ha to 0.26 ha (-82.3%), correcting 1.21 ha of shadows. These were redistributed into Urban (+0.61 ha), Vegetation (+0.61 ha), and Roads (+0.61 ha), while Bare soil decreased by 0.62 ha, reflecting shadows in urban canyons (Table 2).

4. CONCLUSION

- Although the confusion matrix does not reflect changes in the class-wise metrics, the shadow correction did spatially modify the map. From a hydrological perspective, the correction is significant as it reduces the overestimation of water bodies by 17.8% and increases the actual impervious surface area.

ACKNOWLEDGEMENTS

The authors acknowledge the National Council of Science and Technology (CONACYT) for funding through the Doctoral Program in Engineering Sciences.

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